

SYNOPSIS

- 1 What is a Cell?
- 2 Cell Theory
- 3 An Overview of Cell
- 4 Prokaryotic Cells
- 5 Eukaryotic Cells

INTRODUCTION

When you look around, you see both living and non-living things. You must have wondered and asked yourself – ‘what is it that makes an organism living, or what is it that an inanimate thing does not have which a living thing has?’

The answer to this is the presence of the **basic unit of life – the cell** in all living organisms.

All organisms are composed of cells. Some are composed of a single cell and are called **unicellular organisms** while others, like us, composed of many cells, are called **multicellular organisms**.

WHAT IS A CELL?

- **Cell is a fundamental, structural and functional unit of life (all living organisms).**

NOTE: - Unicellular organisms are capable of performing

- (i) Independent existence and
- (ii) Performing the essential function of life.

- **Anton von Leeuwenhoek** first described the **live cells**.
- **Robert hook** first termed the name **CELL**.
- **Robert brown discovered the Nucleus**,

- ◆ The invention of the microscope and its improvement leading to the **electron microscope** revealed all the structural details of the cell.

CELL THEORY:

- Cell theory is put forth by **Mathias Schleiden, Theodore Schwann** together but later it is modified by **Rudolf Virchow**. The cell theory states that,
 - All living organisms are composed of cells and products of cells.
 - All cells arise from pre-existing cells.
 - An organism may be made up of single cell or many cells.

NOTE: - Cells differ greatly in size, shape and activities

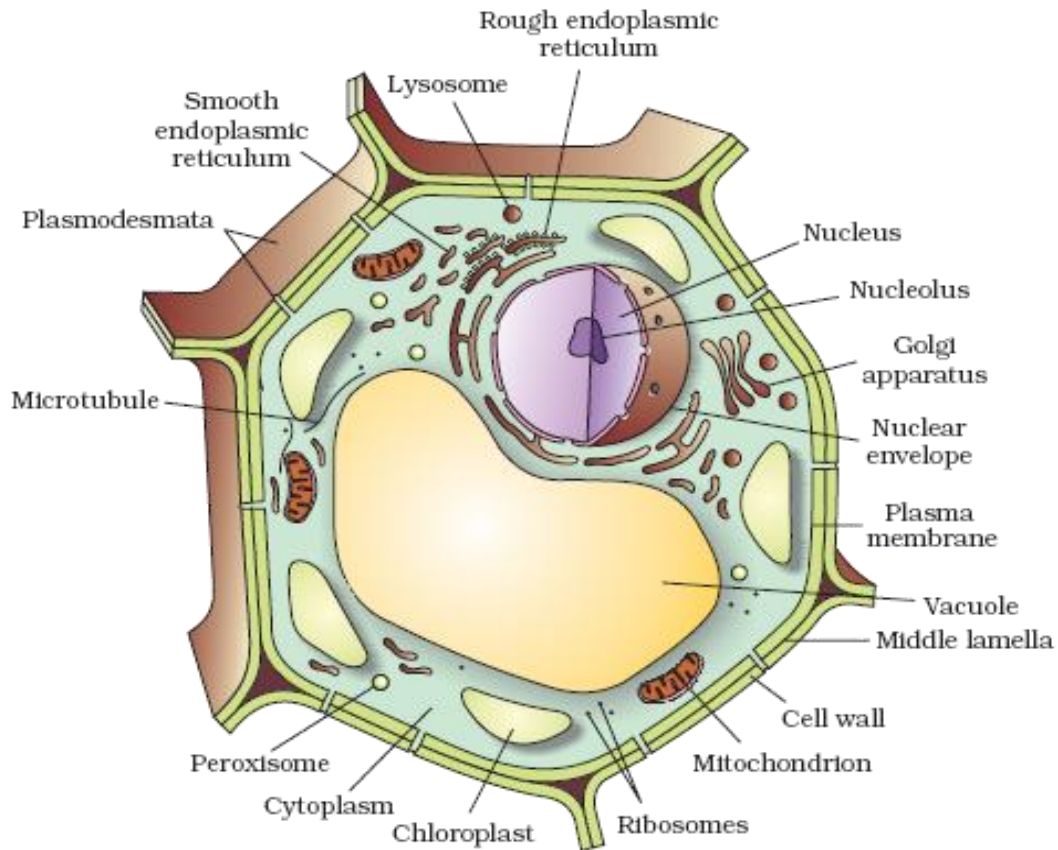
- Smallest cell: **Mycoplasma** (0.3 μm)
- Largest cell: **Ostrich egg**
- Longest cell: **Nerve cell**

AN OVERVIEW OF A CELL

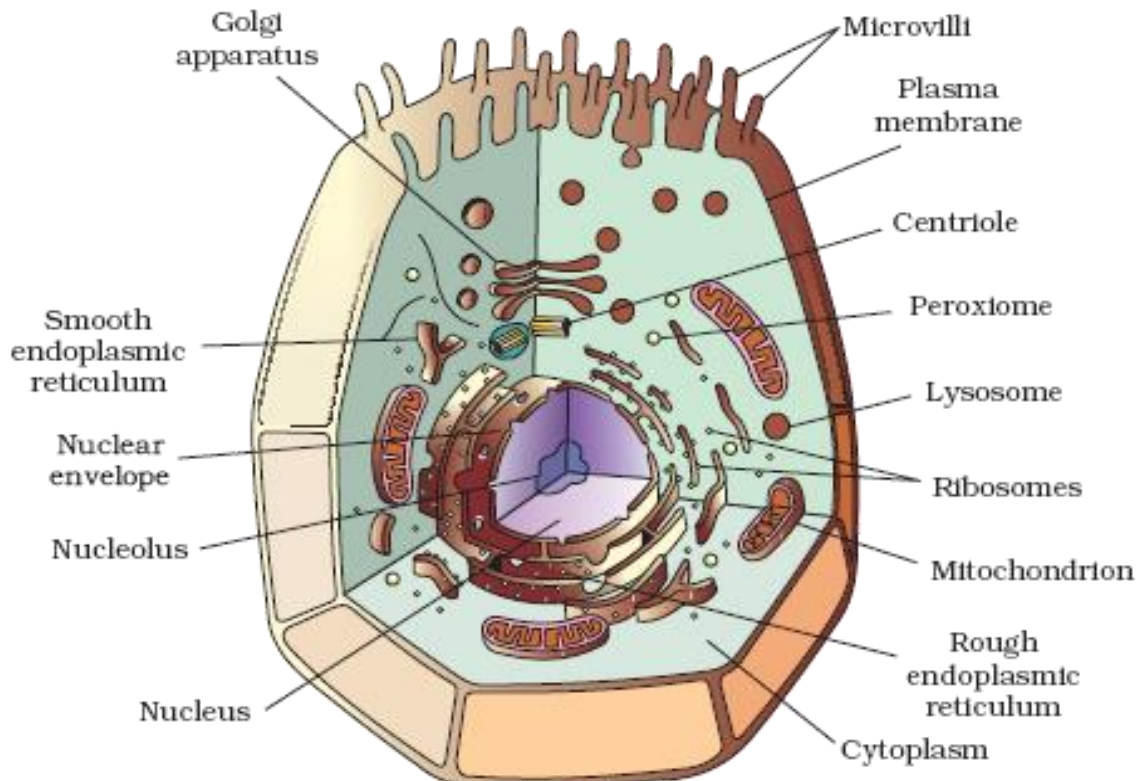
- Every cell is made up of three main components namely **Plasma membrane, cytoplasm, and nucleus**.
- **Cell wall:** - is a distinct outer boundary present in **plant cell**. But it is absent in animal cell.
- **Plasma membrane:** - lies just below the cell wall in plant cells but in animal cells, plasma membrane forms outer most layer.
- **Cytoplasm:** - is a **semi-fluid matrix** which occupies the volume of the cell and contains cell organelles.
 - Cytoplasm is the main region of cellular activities in both the plants and animals, where various bio-chemical reactions occur in it to keep the cell in the “**Living state**”
- **Cell Organelles:** - these are certain distinct **membrane bound** and **non-membrane bound** structures present in the cell cytoplasm.
 - i) **Membrane bound organelles:** - membrane bound organelles present in the cytoplasm are,
 - **Nucleus,**
 - **Endoplasmic reticulum (ER),**
 - **Golgi complex / Golgi apparatus / Golgi bodies,**
 - **Lysosomes,**
 - **Mitochondria,**
 - **Chloroplasts,**
 - **Microbodies and**
 - **Vacuoles.**
 - ii) **Non-membrane organelles:** - non-membrane bound organelles present in the cytoplasm are,
 - **Ribosome** (also present in mitochondria, chloroplast and on the ER)
 - **Centrioles** (present only in animal cells which helps in cell division)
- **Nucleus:** - is a dense **membrane bound structure** present in the cytoplasm of both plant and animal cell. The membrane of nucleus is called **nuclear membrane**. Nucleus contains the **chromosomes** which in turn contain the genetic material, **DNA**.

NOTE: - Based on the presence or absence of the nuclear membrane, cells are classified into two major types; **Prokaryotes** and **Eukaryotes**.

- i) **Prokaryotes:** - The cells which do not have membrane (nuclear membrane) bound nucleus are called prokaryotic cells and the organisms which are made up of prokaryotic cells are called Prokaryotes.
 - Prokaryotes lack membrane bound organelles but possess one non-membrane bound organelle i.e., **Ribosome**.
- ii) **Eukaryotes:** - The cells that have membrane bound nuclei are called eukaryotic cells and the organism which are made up of eukaryotic cells are called eukaryotes
 - Eukaryotes also have other membrane bound organelles beside the nucleus.



PLANT CELL



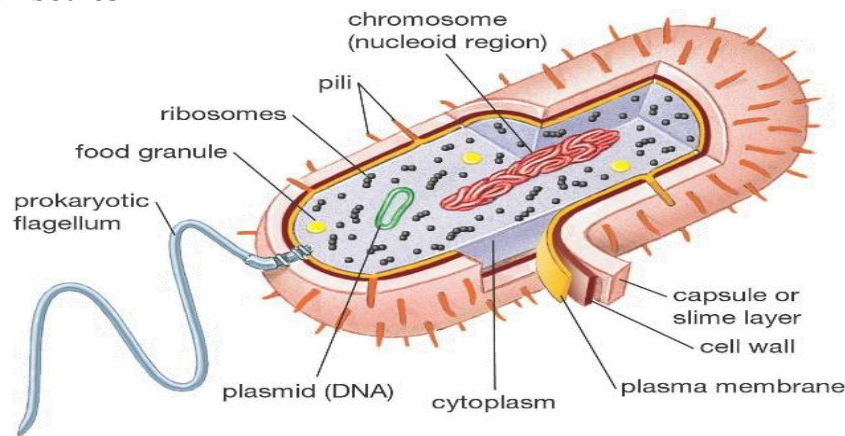
ANIMAL CELL

PROKARYOTIC CELLS

- Prokaryotic cells are represented by **bacteria**, **blue-green algae**, **PPLO** (Pleuro Pneumonia Like Organisms) and **Mycoplasma**.
- They are generally smaller and rapidly multiplying.
- Prokaryotic cells vary greatly in shape and size.

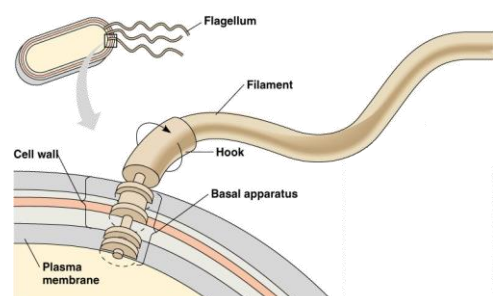
Characteristic features of prokaryotic cells:

- Prokaryotic cells have **cell wall** surrounding the cell membrane.
- Absence of a well-defined nucleus.
- May have **plasmids** – are small **non-genomic DNAs** present in addition to genomic DNAs; this confers characteristics like **antibiotic resistance** to bacteria, and help in **transformation**.
- Absence of membrane bound organelles (only **ribosomes** are found in prokaryotes).
- Have **mesosomes** (specialised differentiated cell membranes); these are infoldings of the cell membrane.
- Have **inclusion bodies**.



Cell Envelope and its modifications in Prokaryotes

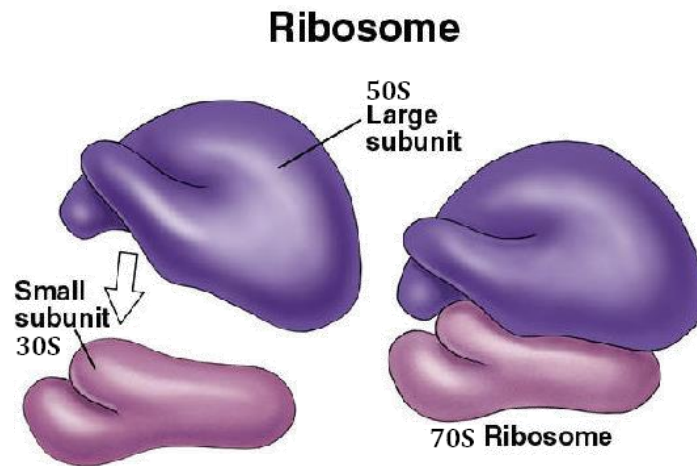
- Cell Envelope** – In prokaryotes cell envelope is **Three-Layered Structure**
 - Glycocalyx** (Outermost): May be present in the form of loose sheath called **slime layer** in some bacteria, or as a thick and tough **capsule** in others.
 - Cell wall** (middle): Determines the shape of a cell and provides a strong structural support.
 - Plasma Membrane** (innermost): is **semi-permeable** and structurally similar to that of eukaryotes.
- Mesosome**
 - Mesosomes are formed by the extension of the plasma membrane into the cell
 - These extensions are in the form of **vesicles, tubules and lamellae**.
 - Functions: Cell wall formation; DNA replication and distribution; respiration and secretion processes; increase surface area of plasma membrane and enzymatic content.
- Prokaryotic Cell may be – Motile or Non-Motile**
 - Motile: Have flagella
 - Non-motile: Lack flagella
- Flagella** have three parts: **Filament, Hook, Basal body**.



- **Pili** (tubular structures made of proteins) and **fimbriae** (bristle-like fibres) are also present along with flagella, but their function is attachment (to the substratum or the host cell).
- Depending upon the cell envelope, bacteria are of **two types**:
 - **Gram positive bacteria** – Bacteria which take up gram stain are called Gram positive bacteria.
 - **Gram negative bacteria** – Bacteria which do not take up gram stain are called Gram negative bacteria.

Ribosome

- Associated with plasma membrane of size 15nm by 20nm.
- Made up of two subunits: **50S** and **30S = 70S**.
- Ribosomes are the **site of protein synthesis**.
- **Polysome**: When several ribosomes attach to a single mRNA is called polysome.
- Ribosome helps in the translation of mRNA into proteins.

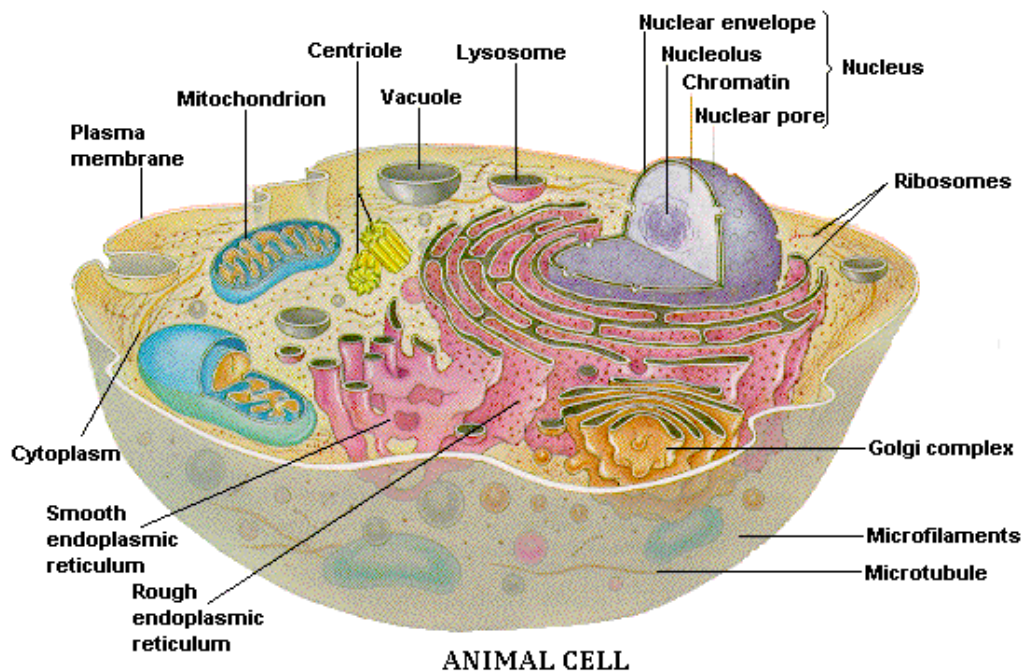
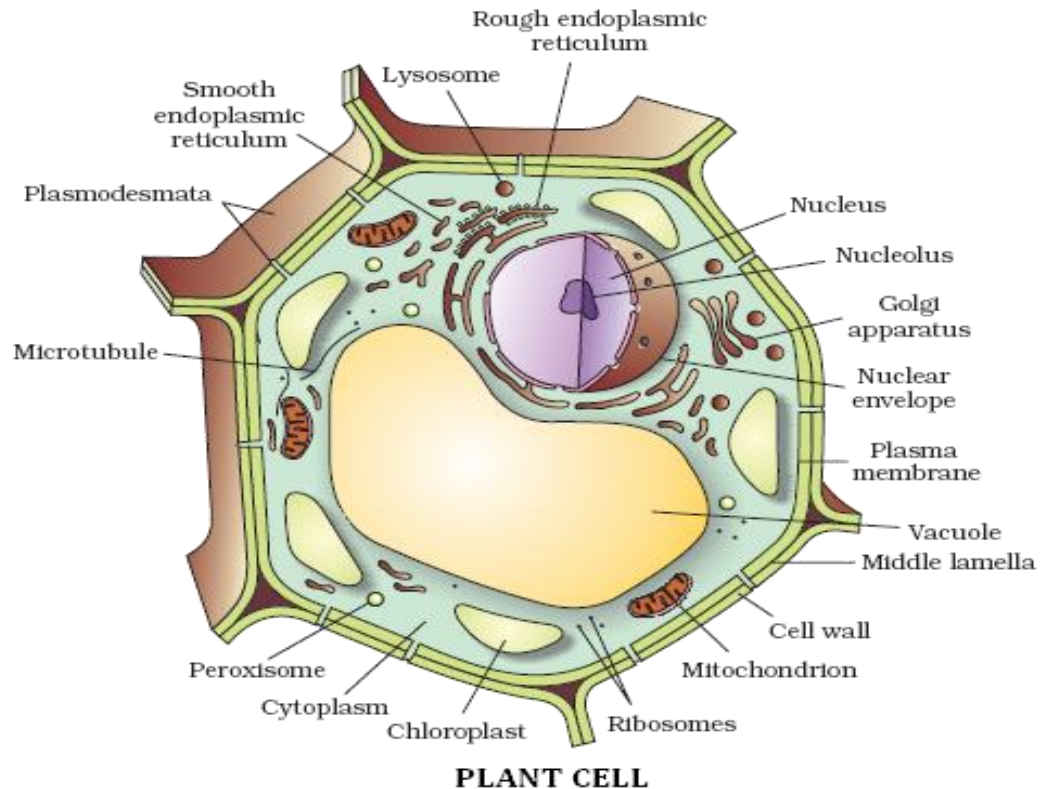


Inclusion Bodies

- Reserve material in prokaryotic cell is stored in the form of inclusion bodies.
- Suspended freely in the cytoplasm, e.g., **phosphate granules**, **glycogen granules** and **cyanophycean granules**.
- **Gas vacuoles**: Inclusion bodies found in blue-green, purple and green bacteria.

EUKARYOTIC CELL

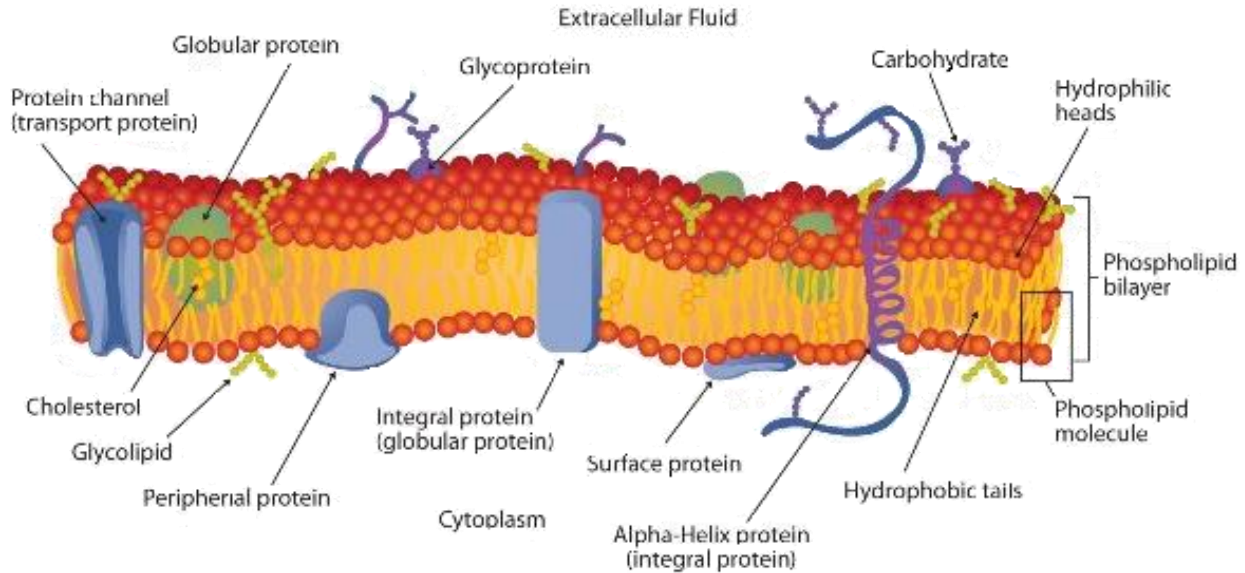
- Eukaryotic cell includes all the protists, plants, animals and fungi.
- Several membrane-bound organelles are present.
- Nucleus has a **nuclear envelope**; genetic material is organised into **chromosomes**.
- Have complex locomotory and cytoskeleton structures.
- Plant cells possess **cell wall**, **plastids** and a **large central vacuole**, which are absent from animal cells.
- Animal cells possess **centrioles**, **lysosomes**, numerous **small-sized vacuoles**, which are absent from plant cells.



Cell Membrane (Plasma Membrane)

- Composed of a **lipid bilayer**, with a polar head towards the outer side, and hydrophobic tails towards the inner side. So, the non-polar, hydrophobic tail is protected from aqueous environment.
- **Proteins** and **carbohydrates** are also present in the cell membrane.
- **Membrane proteins** can be **integral** (lying buried in the membrane) or **peripheral** (lying on the surface of the membrane).

- **Fluid mosaic model** (Singer and Nicolson) shows that the **quasi-fluid nature of lipids** enables lateral movement of proteins within the bilayer.
 - Fluidity is the measure of the ability of the membrane to allow movement within itself.
 - Fluid nature of the membrane is important for cell division, cell growth, the formation of intercellular junctions, secretion and endocytosis.



Fluid mosaic model of plasma membrane

- **Transport across plasma membrane:**
 - **Passive transport** – does not require energy; neutral solute moves by **diffusion**, along the concentration gradient; Water moves through **osmosis**; polar molecules that cannot diffuse move through **facilitated diffusion**, carried out by carrier proteins.
 - **Active transport** – requires energy for transport **against** the concentration gradient.

Cell Wall

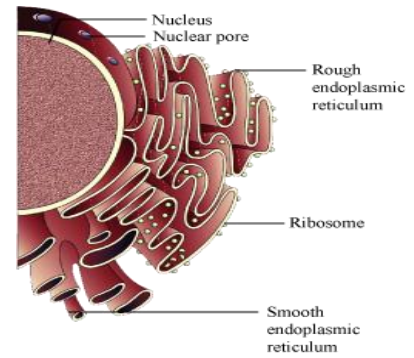
- Present exterior to the cell membrane in **fungi** and **plants**.
- Cell wall gives **shape** and **protection** to cell.
- Helps in cell to cell interactions; provides a **barrier** against undesirable macromolecules
- **Algae cell wall:** Has cellulose, galactans, mannans, minerals (CaCO_3).
- **Plant cell wall:** Has cellulose, hemicelluloses, pectins and proteins.
- In a young plant cell, the primary cell wall is present; it diminishes as the plant matures and is replaced by the secondary cell wall, towards the inner side of the cell.
- Neighbouring plant cells are separated by a layer called **middle lamella**.
- Plasmodesmata traverse the cell wall and the middle lamella, and connect cytoplasm of the neighbouring cells.

ENDOMEMBRANE SYSTEM

- Some organelles with coordinated functions together constitute the endomembrane system.
- Endomembrane system includes the **endoplasmic reticulum (ER)**, **Golgi complex**, **Lysosomes** and **vacuole**.

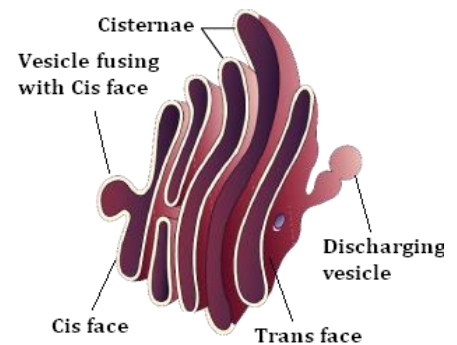
1. Endoplasmic Reticulum (ER):

- ER is Reticulum (network) of tiny tubular structures scattered in the cytoplasm
- ER divides the intracellular space into luminal (inside ER) and extra-luminal (cytoplasm compartments)
- ER is of **two types** – **Rough ER** (has ribosomes attached to its outer surface) and **smooth ER** (ribosomes absent, so it appears smooth)
 - RER is found in cells involved in **proteins synthesis**.
 - SER is the major site for **lipid synthesis**.



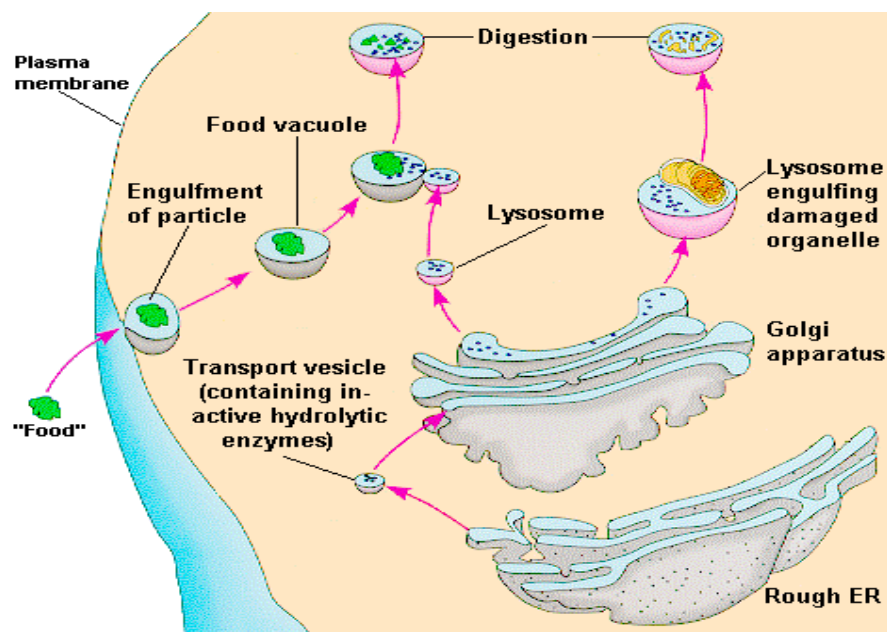
2. Golgi Apparatus:

- Reticular structure consisting of many disc-shaped sacs called **cisternae**, stacked parallel to each other
 - Cisternae have 2 distinct faces – **Convex (cis)** – forming face, **Concave (trans)** – maturing face. *Cis* and *trans* faces are interconnected.
 - Function – packaging of materials, either to be delivered to the intracellular targets or to be secreted outside the cell
 - To be packaged, materials fuse with the *cis* face and move towards the *Trans* face.
 - Glycolipids and glycoproteins are also synthesised in the Golgi apparatus.



3. Lysosomes:

- Lysosomes are vesicles formed as a result of the packaging in the Golgi apparatus.
- These are rich in all types of **hydrolytic enzymes** (acidic), and are capable of digesting all types of molecules, such as carbohydrates, lipids, proteins, nucleic acid.

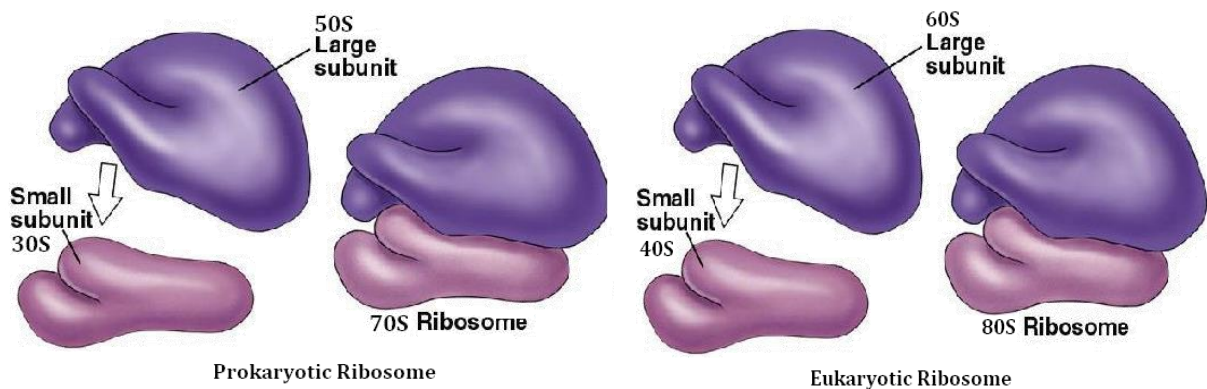


4. Vacuole:

- Membrane-bound space in cells; containing water sap, excretory products and other waste materials
- Vacuole membrane is called as **Tonoplast**.
- Tonoplast has the ability to pump ions into the vacuole against the concentration gradient. So, the concentrations of ions in the vacuole are greater than in the cytoplasm.
- E.g., contractile vacuole is found in *Amoeba* for excretion; food vacuole is found in protists for engulfment of food particles

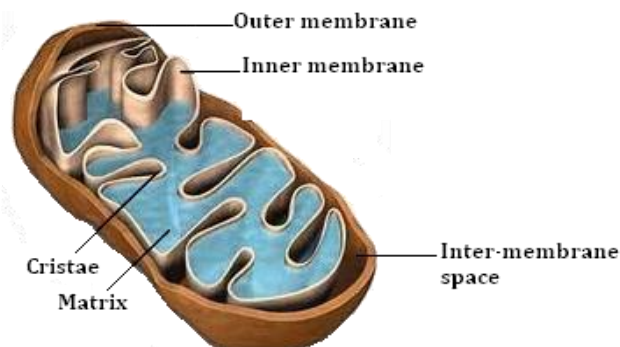
Ribosomes (Site of protein synthesis)

- Granular structures, first observed by **George Palade** under an electron microscope.
- Ribosome is made up of **RNA** and **proteins**.
- Not surrounded by any membrane.
- Prokaryotic cells have ribosomes **70S**. Eukaryotic ribosomes are **80S**. "S" stands for **Svedberg unit**, and is indicative of density and size.



Mitochondria (Power house of cell)

- Mitochondria (Sing.: Mitochondrion) is sausage-shaped or cylindrical cell organelles, visible only with specific staining under the microscope
- Have **double-membrane system**: Inner membrane and Outer membrane
 - These membranes divide the lumen of the mitochondria into **two compartments** – the outer compartment and the inner compartment (filled with **matrix**)
 - **Outer membrane** – forms the boundary of the organelle
 - **Inner membrane** – forms the infoldings called **cristae**; this increases the surface area of the inner membrane
 - Both membranes have specific enzymes associated with them.

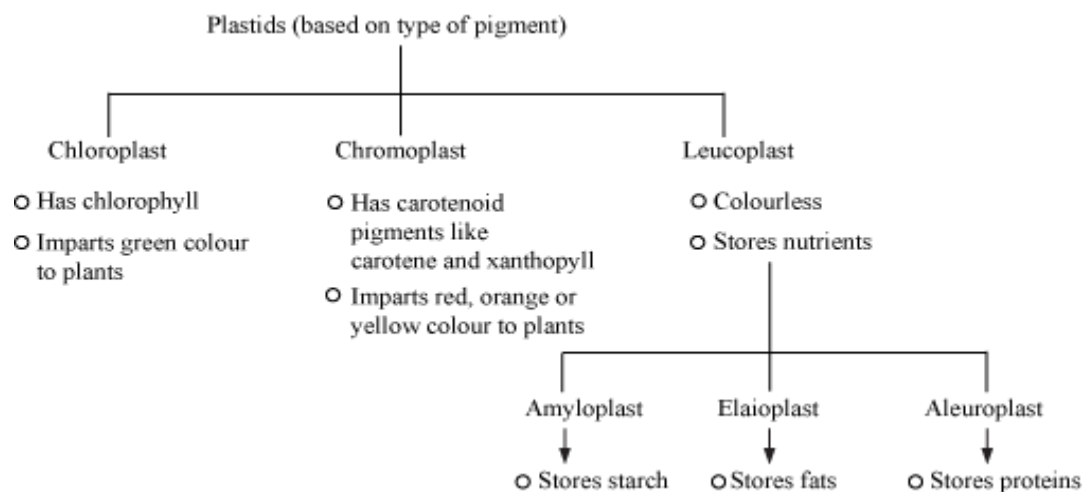


Structure of mitochondrion (Longitudinal section)

- Importance: Mitochondria are called **the 'power houses of cells'** since they are the site for aerobic respiration, and form **ATP** (source of cellular energy).
- Matrix contains components like the **single circular DNA, RNA, ribosomes (70S)** and the components needed for protein synthesis.
- Mitochondria divide by **fission**.

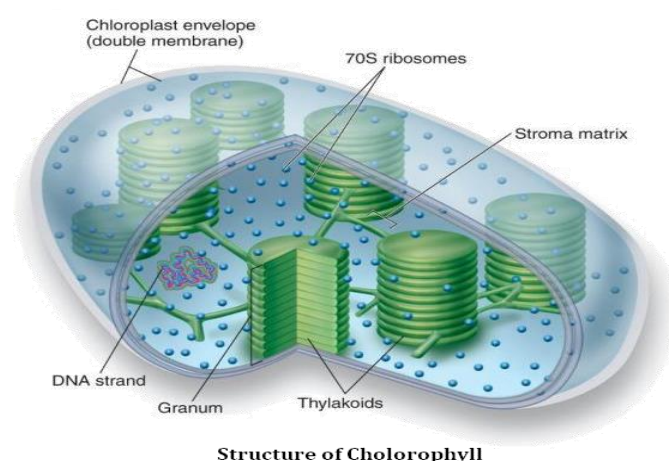
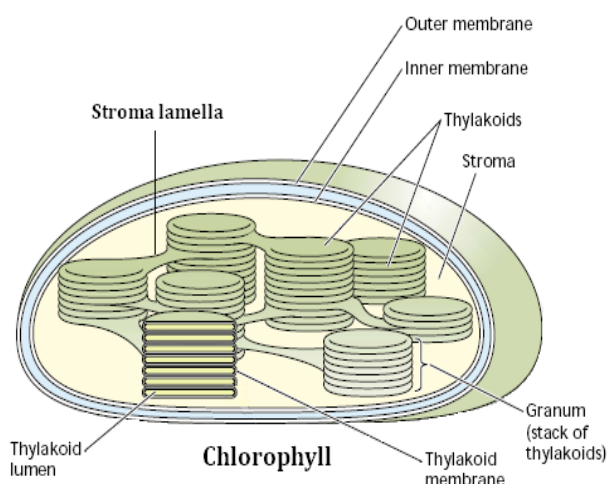
Plastids

- Plastids are found in **plant cells** and **euglenoids**.
- Bear **pigments** and impart **colours** to plants
- Plastids are large-sized, and hence, easily observable.



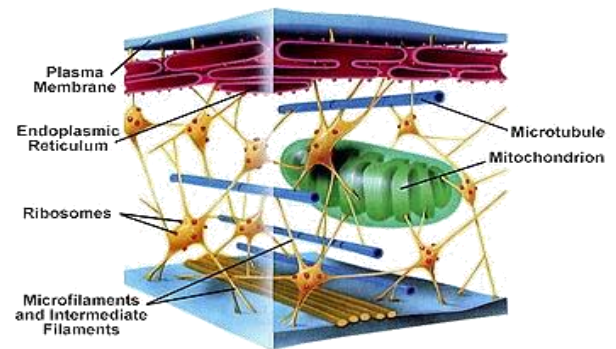
Chloroplast

- Chloroplast are found in **mesophyll cells** in leaves
- Chloroplast can be variable in shape (oval/spherical/discoid/ribbon-like), in length and in number (1 per cell in *Chlamydomonas* to 20–40 per cell in green algae)
- Chloroplast have **double membrane**: Outer membrane and Inner membrane (less permeable)
- **Stroma**: Space enclosed by the inner membrane; contains enzymes required for carbohydrate and protein syntheses
- **Thylakoids**: Flattened, membrane-bound sacs present in the stroma; contains chlorophyll pigment
- **Grana**: Integrated thylakoids; like piles of coins, where thylakoids are arranged in stacks
- **Stroma Lamella**: Flat, membrane-bound tubules; connect thylakoids present in the different grana
- **Lumen**: Space present in thylakoid, i.e., the space enclosed by the thylakoid membrane
- Stroma of the chloroplast has small, **circular DNA** and **ribosomes (70s)**.



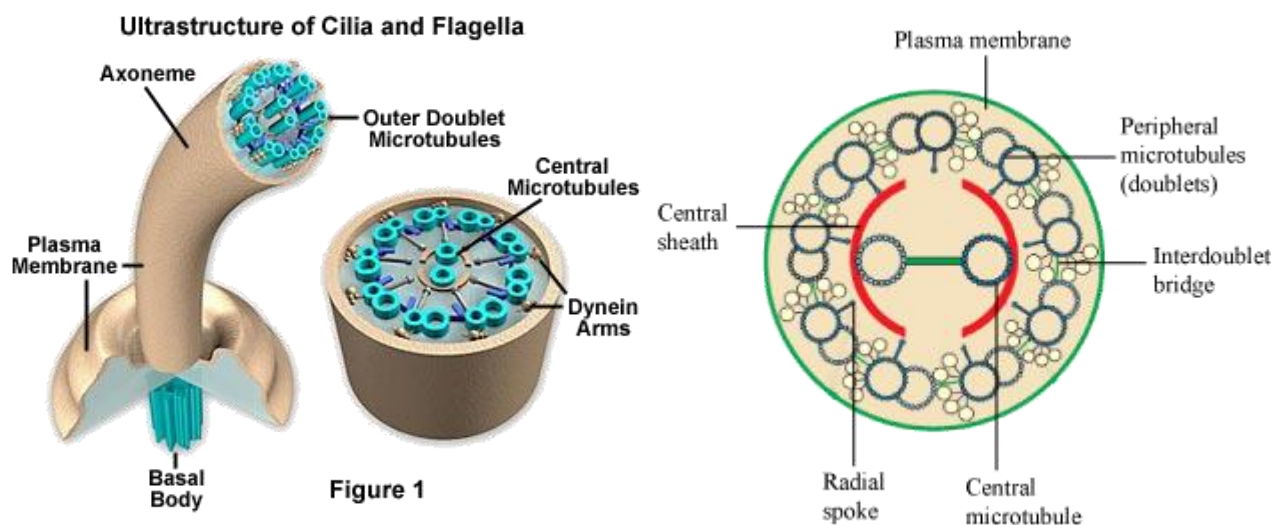
Cytoskeleton

- Filamentous, proteinaceous structures present in cytoplasm
- Functions:
 - Mechanical support
 - Motility
 - Maintenance of the shape of a cell



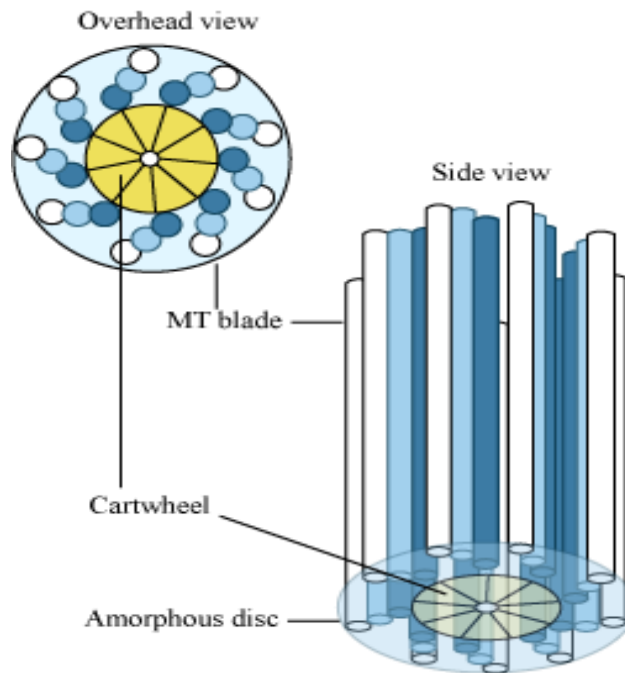
Cilia and Flagella

- These are hair-like outgrowths of the plasma membrane.
- Cilia:** Structures working as oars; help in motility
- Flagella:** Comparatively longer and fewer than the cilia
- Electron microscope study of the cilia and the flagella:
 - They are covered by the plasma membrane.
 - Axoneme** – core of the internal portions of the cilia and the flagella
 - Axoneme contains microtubules. There are **9** pairs of peripheral microtubules and one pair of central microtubules.
 - This arrangement is called **9 + 2 array**.
 - Central tubules** – connected by a bridge and enclosed within the central sheath
 - Radial spoke** – connects one of the tubules of each of the nine peripheral microtubules with the central sheath
 - Linkers** – interconnect the peripheral tubules
 - Basal body – centriole-like structure from where the cilia and the flagella emerge



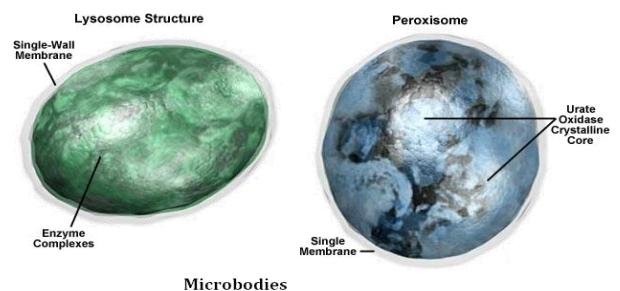
Centrosome and Centrioles

- Centrosome:** Organelle that contains **two** cylindrical structures called **centrioles**.
- Both centrioles in a centrosome lie perpendicular to each other. Centrioles have a cartwheel-like organisation.
- A centriole is made up of **9 peripheral fibrils** of **tubulin** protein. Also, each of the peripheral fibrils is a **triplet**
- Hub:** Central part of the proximal region of a centriole.
- Spokes:** Connect the hub with the tubules of the triplets.
- Importance of centriole: Forms the **basal body** of the cilia and the flagella, and the spindle fibre

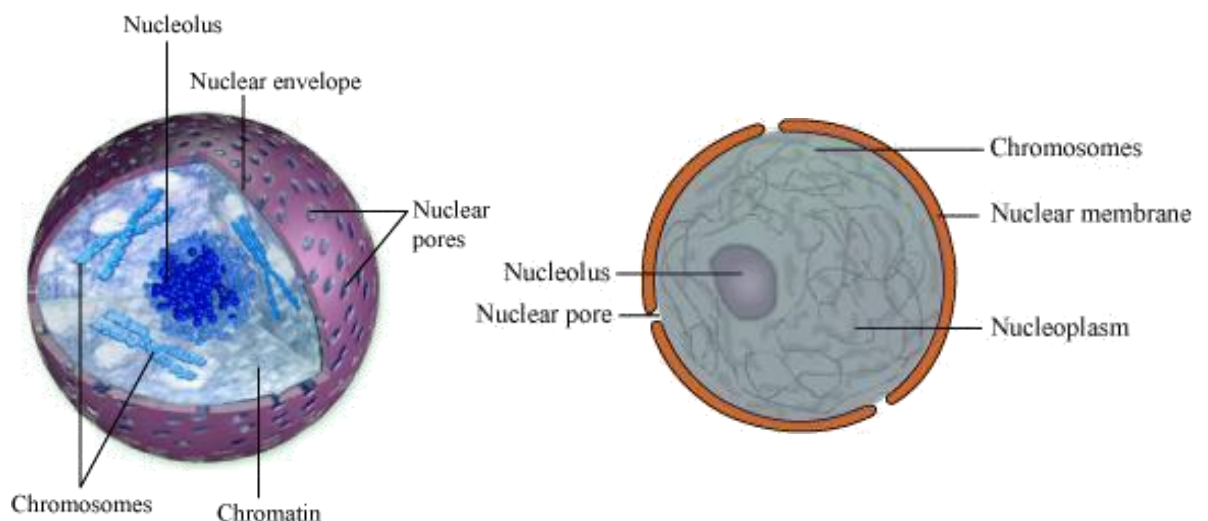


Microbodies

- Microbodies are membrane-bound **minute vesicles**
- Present in both plant and animal cells.
- Contain **enzymes**



Nucleus



- Every cell has a nucleus, except some such as the RBCs of mammals and the sieve tube cells in vascular plants.
- A cell usually has one nucleus, except some variations.
- Nucleus is bound by a **nuclear envelope** which consists of **two membranes** with **perinuclear space** (10 – 50 nm) between them
- Perinuclear space acts as a barrier for the flow of materials between the inside of the nucleus and the cytoplasm. So, to facilitate the transfer of RNA and proteins, **nuclear pores** are present. Nuclear pores are formed at places where the two membranes fuse.

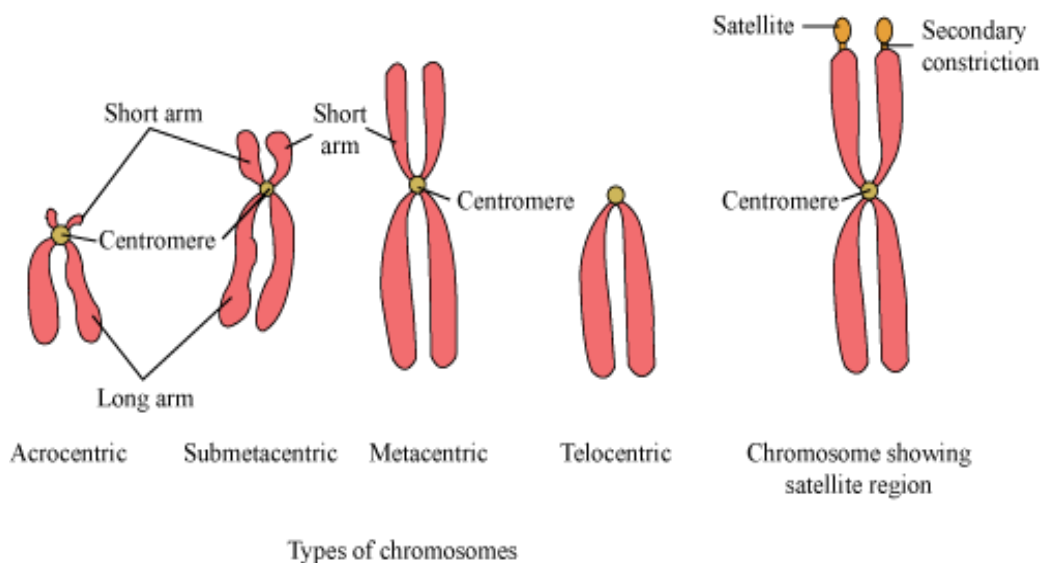
- Nuclear matrix is called **Nucleoplasm** which consists of the **nucleolus** and the **chromatin**.
- **Nucleolus** (pl. nucleoli): Site of active **rRNA synthesis**; not bound by membrane; its contents are continuous with the nucleoplasm
- **Chromatin**: Loose, indistinct network of **nucleoprotein** fibres (present in the interphase)
- **Chromosomes**: Chromatin structures develop into distinct chromosomes during cell division.
 - Chromosome contains **DNA, histone proteins, non-histone proteins**, and also some **RNA**
 - DNA is distributed among **23 pairs (46)** of chromosomes
 - A chromosome has a primary constriction called **Centromere**. On the sides of the centromere, disc-shaped **kinetochores** are present.



Chromosome with kinetochore

Classification of Chromosomes

- Based upon the position of the centromere, chromosomes are of four types:
 - 1 **Metacentric** – centromere located in the middle, forming two equal arms of the chromosome.
 - 2 **Submetacentric** – centromere located slightly away from the middle, resulting in one arm being longer than the other.
 - 3 **Acrocentric** – centromere located close to the end, resulting in one arm being extremely longer than the other.
 - 4 **Telocentric** – centromere located at the terminal point.



- **Satellites**: Small fragments that appear due to the non-staining secondary constrictions present at constant locations on the chromosomes

SUMMARY

All organisms are made of cells or aggregates of cells. Cells vary in their shape, size and activities/functions. Based on the presence or absence of a membrane bound nucleus and other organelles, cells and hence organisms can be named as eukaryotic or prokaryotic.

A typical eukaryotic cell consists of a cell membrane, nucleus and cytoplasm. Plant cells have a cell wall outside the cell membrane. The plasma membrane is selectively permeable and facilitates transport of several molecules. The endomembrane system includes ER, golgi complex, lysosomes and vacuoles. All the cell organelles perform different but specific functions. Centrosome and centriole form the basal body of cilia and flagella that facilitate locomotion. In animal cells, centrioles also form spindle apparatus during cell division. Nucleus contains nucleoli and chromatin network. It not only controls the activities of organelles but also plays a major role in heredity.

Endoplasmic reticulum contains tubules or cisternae. They are of two types: rough and smooth. ER helps in the transport of substances, synthesis of proteins, lipoproteins and glycogen. The golgi body is a membranous organelle composed of flattened sacs. The secretions of cells are packed in them and transported from the cell. Lysosomes are single membrane structures containing enzymes for digestion of all types of macromolecules. Ribosomes are involved in protein synthesis. These occur freely in the cytoplasm or are associated with ER. Mitochondria help in oxidative phosphorylation and generation of adenosine triphosphate. They are bound by double membrane; the outer membrane is smooth and inner one fold into several cristae. Plastids are pigment containing organelles found in plant cells only. In plant cells, chloroplasts are responsible for trapping light energy essential for photosynthesis. The grana, in the plastid, are the site of light reactions and the stroma of dark reactions. The green coloured plastids are chloroplasts, which contain chlorophyll, whereas the other coloured plastids are chromoplasts, which may contain pigments like carotene and xanthophyll. The nucleus is enclosed by nuclear envelop, a double membrane structure with nuclear pores. The inner membrane encloses the nucleoplasm and the chromatin material. Thus, cell is the structural and functional unit of life.